

TECHSPHERE PK

Virtual Internship — Batch 3

Web Development Track

WEEK 2

JavaScript & Responsive Design

Difficulty: (3/5)

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Submit via: Google Classroom Assignment

Program: Summer 2026 · 100% Online · 4 Weeks

Tools: VS Code (free) + Google Chrome + Live Server Extension

📖 Read all instructions carefully before starting. Submit before the deadline. For questions, use the private comment section on Google Classroom.

WEEK 2 — JavaScript & Responsive Design

This week you bring your portfolio page to life with JavaScript interactivity and make it work perfectly on every screen size. These two skills are the difference between a basic webpage and a professional website.

Topics Covered This Week

- JavaScript basics: variables (let, const), functions, if/else conditions
- DOM Manipulation: `document.getElementById`, `querySelector`, `innerHTML`
- Event Listeners: `addEventListener` — responding to clicks, hovers, key presses
- CSS Variables: `--primary-color` syntax — making themes easy to change globally
- `classList` API: `classList.add`, `classList.remove`, `classList.toggle`
- Responsive design: media queries `@media` (`max-width: 768px`)
- Responsive units: % for widths, `rem` for fonts, `vh/vw` for viewport sizing
- Google Fonts: how to embed professional fonts from `fonts.google.com`
- WordPress overview: what it is, when to use it, pages vs posts, themes, plugins

Study Resources

- JavaScript Basics: `javascript.info` — best free JavaScript learning resource
- CSS Media Queries: `css-tricks.com/snippets/css/a-guide-to-flexbox/`
- Google Fonts: `fonts.google.com` — pick any 2 fonts and get the embed code
- WordPress: watch 'WordPress for beginners 2024' on YouTube — 20 minute video

📌 You do not need to learn all of JavaScript this week. Focus only on: variables, functions, `getElementById`, `addEventListener`, and `classList.toggle` — that is enough for this week's task.

✓ YOUR TASK THIS WEEK

Upgrade your Week 1 portfolio page with JavaScript interactivity and complete mobile responsiveness. You are building ON TOP of your Week 1 files — do not start from scratch.

Part A — Dark/Light Mode Toggle (40 points)

Add a button to your portfolio page that switches between dark mode and light mode when clicked.

- Add a toggle button in your navigation bar or header
- Button text: 'Switch to Dark Mode' — changes to 'Switch to Light Mode' after click
- When dark mode is ON: dark background (`#111` or similar) + light text (`#fff`)

- When light mode is ON: white or cream background + dark text (#111 or #333)
- Use CSS Variables for colors so switching is clean and instant

Recommended JavaScript approach:

```
const btn = document.getElementById('themeToggle');
btn.addEventListener('click', function() {
  document.body.classList.toggle('dark-mode');
});
```

Recommended CSS Variables approach:

```
:root { --bg: #ffffff; --text: #111111; }
.dark-mode { --bg: #111111; --text: #ffffff; }
body { background-color: var(--bg); color: var(--text); }
```

Part B — Mobile Responsive Design (35 points)

Your page must work perfectly on a mobile screen (375px width). Test using Chrome DevTools — press F12, then click the phone icon.

- On mobile (375px width): navigation links must stack vertically OR wrap neatly
- On mobile: no horizontal scrolling allowed — everything must fit within the screen
- On mobile: text must remain readable — minimum 16px font size for body text
- On mobile: skills section must stack into 1 or 2 columns maximum
- On mobile: all sections must have proper left and right padding (no text touching edges)

Basic media query template to add in your CSS:

```
@media (max-width: 768px) {
  nav { flex-direction: column; }
  .skills-grid { grid-template-columns: 1fr 1fr; }
  body { padding: 0 16px; }
}
```

Part C — Smooth Scroll Navigation (10 points)

- Make your navigation links scroll smoothly to each section when clicked
- Add this single line to your CSS and it works automatically:

```
html { scroll-behavior: smooth; }
```

- Each section must have an id that matches the navigation href:

```
<section id='about'> ← matches <a href='#about'>
```

Bonus — Google Fonts (optional, +5 points)

- Go to fonts.google.com, pick 2 fonts (one for headings, one for body)
- Click 'Get font' → 'Get embed code' → copy the @import link
- Paste the link in your CSS file at the very top
- Apply the fonts: `body { font-family: 'Your Font', sans-serif; }`

SUBMISSION DETAILS

File 1: Updated ZIP folder with all project files (HTML + CSS + JS)

File 2: Screen recording — 30 to 60 seconds showing:

- : · Dark/Light mode toggle clicking and switching
- : · Mobile responsive view in Chrome DevTools at 375px width

Recording tool: Loom (loom.com — free and easy) OR any screen recorder

Upload: Google Classroom → Week 2 Assignment → upload ZIP + share recording link

Deadline: 7 days from this task being posted

Late policy: 5 points deducted per day late

GRADING BREAKDOWN

- Dark/Light mode toggle works correctly when button is clicked **[40 pts]**
- Page is mobile responsive at 375px — no horizontal scroll, readable text **[35 pts]**
- Smooth scroll navigation working with correct section IDs **[10 pts]**
- Screen recording submitted showing both features working **[15 pts]**

TOTAL: 100 points | PASS MARK: 70/100

 Keep your JavaScript in a separate `script.js` file — link it at the bottom of your HTML just before the closing `</body>` tag. This is best practice and keeps your code organized.

Mentor Tips:

- Use `console.log()` inside your JavaScript to test if your code is running — check DevTools Console tab
- In Wireshark... wait, wrong track! In Chrome DevTools → Console tab — errors appear in red, read them carefully
- CSS Variables are one of the most powerful CSS features — learn them well this week
- If your toggle is not working, check: is the button ID in HTML matching `getElementById('correctID')`?
- Mobile responsiveness test: slowly drag the DevTools panel wider/narrower to see how your layout behaves
- Ask for help early — do not wait until day 6 if you are stuck on day 2

